



KTU NOTES

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**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

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3. A thyristor can be triggered by an external gate pulse - Justify using analogy of thyristor
4. Explain how the firing angle of an SCR can be varied by using a UJT oscillator
5. Two thyristors having a difference of 4 mA in latching current are connected in series. The voltage across the devices are 500 V and 480 V. Calculate the value of the static equalizing resistance value for maximum string efficiency
6. Define holding current and latching current of SCR. Show these curves on the IV characteristics of SCR
7. With the help of circuit diagram and waveform, explain the operation of RC triggering circuit for one thyristor.
8. Discuss the condition which must be satisfied for turning on the SCR by a gate signal
9. Explain the significance of di/dt protection in thyristors and describe the circuit employed for improving the same.
10. Describe briefly the RC triggering circuit for SCR with a neat circuit diagram. With the help of a graph explain how firing angle control up to 180 degrees is achieved.
11. Compare the characteristic features of MOSFET AND IGBT
12. Draw static characteristics of SCR, and based on that explain different modes of operation of SCR
13. With the help of circuit diagram and waveform explain the operation of RC triggering circuit for one thyristor
14. With the help of circuit diagram and waveform explain the operation of RC triggering circuits for one SCR and also draw the voltage across the SCR.
15. Differentiate between features of UJT firing circuit, RC triggering circuit and RC triggering circuit.
16. Mention the important ratings of the thyristors along with their significance
17. A battery is charged through a single phase half wave controlled rectifier. The supply voltage is 230 V, 50 Hz and battery emf is constant at 160 V. Calculate the average charging current for firing angle of 30 degrees. Internal resistance of battery is $2\ \Omega$.
18. Explain R firing circuit of SCR with circuit diagram and waveform

and holding current

3. Compare Thyristor, Power MOSFET and IGBT on the basis of following:
i) Switching frequency ii) Voltage and current ratings iii) Applications
4. Explain with figures the switching characteristics of SCR during turn-on and turn-off.
5. With help of detailed structure explain the operation of MOSFET.
6. Deduce the Two Transistor Model for a Thyristor and explain the operation of Thyristor using this model.
7. Describe the variation of current and voltage during turn-on time of SCR with the help of characteristics.
8. With circuit diagram and relevant waveforms, explain the operation of a triggering circuit for triggering a SCR.